

STS 4600: The Engineer, Ethics, and Professional Responsibility

STS Research Paper

Research Project

Imagine that an interdisciplinary journal of undergraduate research published by UVa¹ is soliciting submissions from engineering students that consider the intersection of technology, society, and ethics. In a research paper of 3000-3750 words (about 12-15 pages), develop an argument that draws on a scholarly STS or ethical framework to analyze the social and/or ethical impacts, implications, or aspects of a particular technology of your choosing as illustrated in a specific case (i.e., incident, example).

Make sure your argument focuses on a specific case (i.e., incident, example) for your analysis and not a general topic / technology (e.g., the Apollo 1 command module fire – not the Cold War era space race, the Apollo program, or NASA spacecraft generally).

Your argument about the technology featured in the case should draw on one specific scholarly STS or ethical framework (e.g., actor-network theory, virtue ethics, etc.). The framework will supply both the research question you ask of the case and the organization of the main points in your argument.

To support your argument, draw on information from at least 10 sources (primary and secondary); these sources should be cited in both the paper (in-text) and the References list at the end (full citation) in APA style.

The primary audience for your paper is UVa undergraduates who may or may not be in your field or in SEAS (i.e., educated, though non-expert, peer readers).

The paper may focus on the technology featured in the STS Project proposal of the Prospectus or on another technology of your choosing. That is, you do not have to write the STS 4600 Research Paper on the same STS topic you proposed in the STS 4500 Prospectus.

Structure

Your research paper should contain the following six sections in order (additional sections are optional, as needed, depending on the nature of the project). These sections are genre conventions common to academic research essays in most fields and, as such, correspond to reader expectations.

1. Introduction (problem frame) – includes the paper's main claim at the end
 - *Background (optional) – provides additional information on the case and/or technology
2. Literature Review – discusses at least two peer-reviewed academic sources related to your research question
3. Conceptual Framework – identifies and explains aspects of the STS/ethical framework you deploy to organize your argument
 - *Method (optional) – Describes the method(s) used to obtain and/or analyze evidence

4. Analysis – explicitly applies the STS/ethical framework to analyze primary source evidence that supports your claim
5. Conclusion – summarizes the argument and indicates its significance to the paper’s audience
6. References list, formatted in APA style – not included in the word/page count

In the weeks to come, we will discuss practices that make for successful research, the fundamentals of academic argument, and how to write each section of the research paper (with reference to scholarly and student examples).

Requirements and Formatting

- Length: About 3000-3750 words, excluding references and figures; include a word count on the last page
- Pages: 12-point New Times Roman font, double spacing for all text (including entries in reference lists), 1” margins on all sides, left justification, page numbers on the bottom center of each page
- Prose: Grammatically accurate; formal professional language; American English; clear, accessible, reader-oriented prose
- Headings: Include headings within the text to guide the reader
- Figures: Include figures to support the explanation(s) in the text (as needed)
- Title Page: Include a title page that conforms to the template on Canvas
- References: Include a Reference List with 10-15 sources cited in the text and formatted in APA style (see the APA Style Quick Guide on Canvas or the APA Style section of Purdue University’s Online Writing Lab (OWL) for help with APA formatting of sources:

https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_style_introduction.html

Deliverables (see the Syllabus class schedule for due dates)

- Workshop Draft of STS Research Paper
- STS Research Paper Draft
- STS Research Paper Revision

¹ See: <https://undergraduateresearch.virginia.edu/present-and-publish/undergraduate-symposia>